

Lecture 3. Simple Linear Regression II - Inferences

Phuong L. Vo

International University

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Outline

- 1 Inferences concerning the slope, β_1
- 2 Inferences concerning the intercept, β_0
- 3 Interval estimation of the mean response, $E\{Y_h\}$
- 4 Prediction of new observation
- 5 Confidence band of regression line
- 6 Analysis of variance for regression analysis
 - Sum of squares, mean square, ANOVA table
 - F test of $\beta_1 = 0$ versus $\beta_1 \neq 0$
- 7 Coefficient of Determination
- 8 Regression through the origin
- 9 Considerations in applying regression analysis

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Motivation of inferences about β_1

- The market research analyst studying the relation between sales (Y) and advertising expenditures (X) may wish to obtain an interval estimate of β_1
- Since β_1 provides information like how many additional sales dollars, on the average, are generated by an additional dollar of advertising expenditure.

Properties of b_1

- Least-square estimator b_1 is given by

$$b_1 = \frac{\sum_i^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i^n (X_i - \bar{X})^2} \quad (1)$$

where n is the number of data points.

- b_1 is a linear combination of Y_i

$$b_1 = \sum_i^n k_i Y_i,$$

where

$$k_i = \frac{X_i - \bar{X}}{\sum_i^n (X_i - \bar{X})^2} \quad (2)$$

Sampling distribution of b_1

For normal error regression model $N(0, \sigma^2)$, the sampling distribution of b_1 is normal with mean and variance:

$$E\{b_1\} = \beta_1$$
$$\text{var}\{b_1\} = \frac{\sigma^2}{\sum_i^n (X_i - \bar{X})^2}$$

- Unbiased estimator
- Replacing σ^2 by the unbiased estimator, MSE, we have the unbiased estimator of $\text{var}\{b_1\}$ as follows

$$s^2\{b_1\} = \frac{MSE}{\sum_i^n (X_i - \bar{X})^2}.$$

Confidence interval for β_1

$\frac{b_1 - \beta_1}{s\{b_1\}}$ follows t -distribution with $(n - 2)$ degree of freedom for normal error regression model.

- $1 - \alpha$ confidence interval of β_1 is:

$$b_1 \pm t(1 - \alpha/2; n - 2)s\{b_1\}$$

Example: Toluca company

- Management wishes an estimate of β_1 with 95 percent confidence coefficient.
- Previous results for Toluca Company Example:

$$n = 25, \bar{X} = 70.0, \sum_i^n (X_i - \bar{X})^2 = 19,800$$

$$b_0 = 62.37, b_1 = 3.5702$$

$$\hat{Y} = 62.37 + 3.5702X, \sum_i^n (Y_i - \bar{Y})^2 = 307,203$$

$$SSE = 54,825, MSE = 2,384.$$

Example: Toluca company

- Standard deviation of b_1 :

$$s^2\{b_1\} = \frac{MSE}{\sum(X_i - \bar{X})^2} = 0.12040$$
$$s\{b_1\} = 0.347.$$

- A 95 percent confidence coefficient: $t(.975; 23) = 2.069$.
- Therefore,

$$3.5702 - 2.069(0.3470) \leq \beta_1 \leq 3.5702 + 2.069(.3470)$$
$$2.85 \leq \beta_1 \leq 4.29.$$

- With confidence coefficient .95, we estimate that the mean number of work hours increases by somewhere between 2.85 and 4.29 hours for each additional unit in the lot.

Tests concerning β_1

- Since $(b_1 - \beta_1)/s\{b_1\}$ is distributed as t with $n - 2$ degrees of freedom, tests concerning β_1 can be set up in ordinary fashion using the t distribution.
- Recall the testing related to t distribution: 1-side test, 2-side test.

Toluca company example, two-sided test

- Test whether or not there is a linear association between work hours and lot size, *i.e.*, whether or not $\beta_1 = 0$. Two alternatives are:

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

- The analyst wishes to control the risk of a Type I error at $\alpha = 0.05$. (Actually, H_a can be concluded at once since the calculated confidence interval does not include 0.)

Toluca company example, two-sided test

- The formal test statistic is:

$$t^* = \frac{b_1}{s\{b_1\}}$$

- With the level of significance at α ,
 - If $|t^*| \leq t(1 - \alpha/2; n - 2)$, conclude H_0
 - If $|t^*| > t(1 - \alpha/2; n - 2)$, conclude H_a
- In Toluca example, where $\alpha = 0.05$, $b_1 = 3.5702$, and $s\{b_1\} = 0.3470$, we obtain $t(0.975; 23) = 2.069$. Since, $|t^*| = |3.5702/0.3470| = 10.29 > 2.069$, we conclude H_a .

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Sampling distribution of b_0

Recall that $b_0 = \bar{Y} - b_1\bar{X}$, which is a linear combination of observation Y_i like b_1 .

For normal error regression model with $\epsilon_i \sim N(0, \sigma^2)$, the sampling distribution of b_0 is normal, with mean and variance are given by:

$$E\{b_0\} = \beta_0,$$
$$\text{var}\{b_0\} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right)$$

An estimator of $\text{var}\{b_0\}$ is obtained by replacing σ^2 with its unbiased estimator, MSE:

$$s^2\{b_0\} = \text{MSE} \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right)$$

Confidence interval for β_0

Similarly to β_1 ,

$\frac{b_0 - \beta_0}{s\{b_0\}}$ follows t -distribution with $(n - 2)$ degree of freedom for normal error regression model.

$1 - \alpha$ confidence limits for β_0 is:

$$b_0 \pm t(1 - \alpha/2; n - 2)s\{b_0\}.$$

Example: calculate the confidence interval of β_0 in Toluca example.

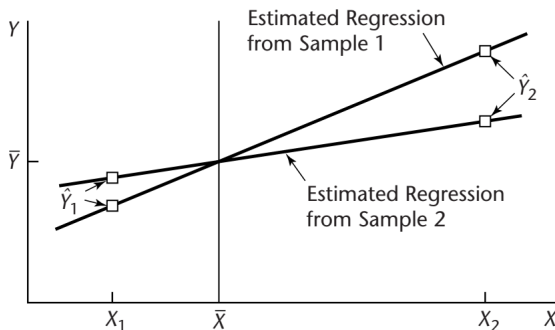
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Interval estimation of $E\{Y_h\}$

- Let X_h denote the level of X for which we wish to estimate the mean response.
- The mean response when $X = X_h$ is denoted by $E\{Y_h\}$.
- Estimator $\hat{Y}_0$ of $E\{Y_h\}$:

$$\hat{Y}_h = b_0 + b_1 X_h.$$



Sampling distribution of $E\{Y_h\}$

For normal error regression model, the sampling distributing of $\hat{Y}_h$ is normal, with mean and variance:

$$E\{\hat{Y}_h\} = E\{Y_h\}$$
$$\text{var}\{\hat{Y}_h\} = \sigma^2 \left(\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2} \right)$$

We can replace σ^2 by its unbiased estimator $MSE = \frac{\sum e_i^2}{n-2}$.

Confidence interval of the mean response, $E\{Y_h\}$

$\frac{\hat{Y}_h - E\{Y_h\}}{s\{\hat{Y}_h\}}$ is distributed as $t(n-2)$ for normal error regression model.

Confident interval of $E\{Y_h\}$: $1 - \alpha$ confidence limits are

$$\hat{Y}_h \pm t(1 - \alpha/2; n - 2)s\{\hat{Y}_h\}.$$

Toluca company example

Find 90 percent confidence interval for $E\{Y_h\}$ when the lot size is $X_h = 65$ units.

- $\hat{Y}_h = 62.37 + 3.5702 * 65 = 294.4$
- $s^2\{Y_h\} = 2,384\left(\frac{1}{25} + \frac{(65-70)^2}{19,800}\right) = 98.37$, hence $s\{Y_h\} = 9.918$.
- confidence interval with confidence coefficient 0.90 is:

$$294.4 - 1.714(9.918) \leq E\{Y_h\} \leq 294.4 + 1.714(9.918)$$
$$277.4 \leq E\{Y_h\} \leq 311.4.$$

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Prediction interval for y_h when parameters known

- Consider the regression relation between the high school grade point average (GPA) of admitted students and the first-year college GPA. The regression model has the parameters:

$$\begin{aligned}\beta_0 &= 0.10, & \beta_1 &= 0.95 \\ E\{Y\} &= 0.10 + 0.95X \\ \sigma &= 0.12.\end{aligned}$$

- The mean college GPA for students whose high school average is $X_h = 3.5$ is: $E\{Y_h\} = 0.10 + 0.95 * 3.5 = 3.425$. The standard deviation is $\sigma = 0.12$ (normal error regression model).
- Suppose we were to predict that the college GPA will be between: $E\{Y_h\} \pm 3\sigma$ (99.7 percent of confident), so that the prediction interval would be: $3.065 \leq Y_h \leq 3.785$.

Prediction Interval for y_h when parameters known

In general, when the regression parameters of normal error regression model are known, the $1 - \alpha$ prediction limits for Y_h are:

$$E\{Y_h\} \pm z(1 - \alpha/2)\sigma.$$

Prediction Interval for y_h when parameters unknown

Prediction limits for y_h must take account of two elements:

- ① Variation in possible location of the distribution of Y (variation of $\hat{Y}_h$).
- ② Variation within the probability distribution of Y .

$\frac{Y_h - \hat{Y}_h}{s_{\{pred\}}}$ is distributed as $t(n - 2)$ for normal error regression model.

Prediction Interval for y_h when parameters unknown

- $\text{var}\{pred\}$: the variance of the prediction error:

$$\begin{aligned}\text{var}\{pred\} &= \text{var}\{Y_h - \hat{Y}_h\} \\ &= \text{var}\{Y_h\} + \text{var}\{\hat{Y}_h\} \\ &= \sigma^2 + \text{var}\{\hat{Y}_h\}\end{aligned}$$

- Note that $\text{var}\{pred\}$ has two components:
 - 1 The variance of the distribution of Y at $X = X_h$,
 - 2 The variance of the sampling distribution of $\hat{Y}_h$.
- Hence, an unbiased estimator of $\text{var}\{pred\}$ is:

$$s^2\{pred\} = MSE + s^2\{\hat{Y}_h\} = MSE \left[1 + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2} \right].$$

Toluca company example

- From previous calculations:
 $Y_h = 419.4, s^2\{\hat{Y}_h\} = 203.72, MSE = 2,384.$
- We obtain $s\{pred\} = 50.87.$
- Hence, the 90 percent prediction interval for Y_h is:

$$419.4 - 1.714(50.87) \leq Y_h \leq 419.4 + 1.714(50.87)$$
$$332.2 \leq Y_h \leq 506.6$$

- This prediction interval is rather wide since MSE is large.

Prediction of mean of m new observations for given X_h

- The $1 - \alpha$ prediction limits are,

$$\hat{Y}_h \pm t(1 - \alpha/2; n - 2)s\{predmean\},$$

where

$$\begin{aligned}s^2\{predmean\} &= \frac{MSE}{m} + s^2\{\hat{Y}_h\} \\ &= MSE \left[\frac{1}{m} + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2} \right].\end{aligned}$$

- Student apply to Toluca company example to predict the mean of work hours per lot with three runs ($m = 3$) and $X_h = 100$.

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Confidence band of regression line

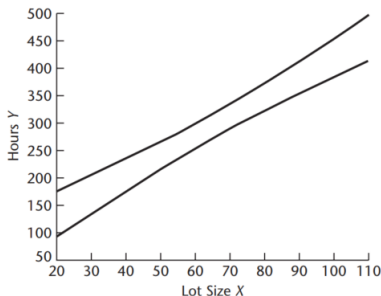
- Working-Hotelling $1 - \alpha$ confidence band

$$\hat{Y}_h \pm Ws\{\hat{Y}_h\},$$

where $W^2 = 2F(1 - \alpha; 2, n - 2)$.

- Toluca example: with $X_h = 100$, $\hat{Y}_h = 419.4$, $s\{\hat{Y}_h\} = 14.27$. Hence, $W^2 = 2F(.90; 2, 23) = 5.098$.

Therefore, confidence band is $387.2 \leq \beta_0 + \beta_1 X_h \leq 451.6$.



Outline

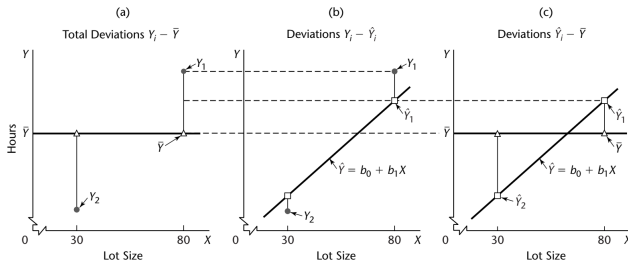
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- We consider the regression analysis from the perspective of analysis of variance
- This new perspective will not enable us to do anything new, but the analysis of variance approach will come into its own when we take up multiple regression models and other types of linear statistical models

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Partitioning of total sum of squares



- The variation measured in terms of the deviations of the Y_i around their mean $\bar{Y}$: $Y_i - \bar{Y}$ (Fig.a)
- Total sum of squares (SSTO, a.k.a. SS_T): a measure of the uncertainty pertaining to the work hours required for a lot, when the lot size is not taken into account.

$$SSTO = \sum (Y_i - \bar{Y})^2$$

Partitioning of total sum of squares

- the variation reflecting the uncertainty concerning the variable Y is that of the Y_i observations around the fitted regression line: (Fig.b)

$$Y_i - \hat{Y}_i.$$

- error sum of squares (SSE, a.k.a. SS_{Res}): The measure of variation in the Y_i observations that is present when the predictor variable X is taken into account is the sum of the squared deviations:

$$SSE = \sum (Y_i - \hat{Y}_i)^2$$

Partitioning of total sum of squares

- The difference between the fitted value on the regression line and the mean of the fitted values $\bar{Y}$ (Fig.c)

$$\hat{Y}_i - \bar{Y}$$

- regression sum of squares (SSR, a.k.a, SS_R):

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2.$$

$$\begin{aligned}
SSTO &= \sum (Y_i - \bar{Y})^2 = \sum [(Y_i - \hat{Y}_i) - (\hat{Y}_i - \bar{Y})]^2 \\
&= \sum (Y_i - \hat{Y}_i)^2 + \sum (\hat{Y}_i - \bar{Y})^2 - 2 \sum (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y}) \\
&= SSE + SSR + 2 \sum \hat{Y}_i(Y_i - \hat{Y}_i) - 2\bar{Y} \sum (Y_i - \bar{Y}) \\
&= SSE + SSR
\end{aligned}$$

The last terms equal to 0 because of the properties of fitted function.

- SSTO has $n - 1$ df,
- SSE has $n - 2$ df (two degrees of freedom are lost because the two parameters β_0 and β_1 are estimated in obtaining the fitted values $\hat{Y}_i$)
- SSR has one degree of freedom associated with it

Mean squares

A sum of squares divided by its associated degrees of freedom is called a mean square (abbreviated MS).

- Regression mean square

$$MSR = \frac{SSR}{1} = SSR$$

- Error mean square

$$MSE = \frac{SSE}{n - 2}$$

- Mean square are not additive: $\frac{SSTO}{n-1} \neq MSR + MSE$.

ANOVA table for simple linear regression

Source of Variation	SS	df	MS	$E\{MS\}$
Regression	$SSR = \sum(\hat{Y}_i - \bar{Y})^2$	1	$MSR = \frac{SSR}{1}$	$\sigma^2 + \beta_1^2 \sum(X_i - \bar{X})^2$
Error	$SSE = \sum(Y_i - \hat{Y}_i)^2$	$n - 2$	$MSE = \frac{SSE}{n - 2}$	σ^2
Total	$SSTO = \sum(Y_i - \bar{Y})^2$	$n - 1$		

- Expected Mean Squares:

$$E\{MSE\} = \sigma^2$$

$$E\{MSR\} = \sigma^2 + \beta_1^2 \sum(X_i - \bar{X})^2$$

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F test of $\beta_1 = 0$ versus $\beta_1 \neq 0$

- The analysis of variance also provides a test for:

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

- $F^* = \frac{MSR}{MSE}$ follows F distribution.
- Test statistic:

$$F^* = \frac{MSR}{MSE}$$

With confident level α :

- If $F^* \leq F(1 - \alpha; 1, n - 2)$, conclude H_0
- If $F^* > F(1 - \alpha; 1, n - 2)$, conclude H_a

where $F(1 - \alpha; 1, n - 2)$ is the $(1 - \alpha)100$ percentile of the distribution.

F test of $\beta_1 = 0$ versus $\beta_1 \neq 0$

Show Toluca example.

Equivalence of F test and t test in simple linear model

$$\begin{aligned} F^* &= \frac{SSR/1}{SSE/(n-2)} = \frac{b_1^2 \sum (X_i - \bar{X})^2}{MSE} \\ &= \frac{b_1^2}{s^2\{b_1\}} = \left(\frac{b_1}{s\{b_1\}} \right)^2 = (t^*)^2 \end{aligned}$$

- In Toluca company example, $F^* = 105.9 = (10.29)^2$, which is $(t^*)^2$.
- The t test is two-tailed whereas the F test is one-tailed. Hence, t test is more flexible since it can be used for one-sided alternatives, while the F test cannot.

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Coefficient of Determination

Coefficient of determination:

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

- $0 \leq R^2 \leq 1$
- Percentage of reduction of total variation associated with the use of the predictor variable X .
- The larger R^2 is, the more the total variation of Y is reduced by introducing the predictor variable X .

Coefficient of Determination

- 1 For the Toluca Company example, we obtained $SSTO = 307,203$ and $SSR = 252,378$. Hence:

$$R^2 = \frac{252,378}{307,203}$$

- 2 Thus, the variation in work hours is reduced by 82.2 percent when lot size is considered.

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Regression through the origin

The no-intercept regression model:

$$y = \beta_1 x + \epsilon$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2}$$

$$s\{\sigma^2\} = MSE = \frac{\sum_{i=1}^n y_i^2 - \hat{\beta}_1 \sum_{i=1}^n y_i x_i}{n - 1}$$

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Considerations in applying regression analysis

- The validity of the regression application depends upon whether basic causal conditions in the period ahead will be similar to those in existence during the period upon which the regression analysis is based.
- In predicting new observations on Y , the predictor variable X often has to be predicted.
- If the levels of the predictor variable that fall outside the range of observations, extreme caution should be exercised.

Considerations in applying regression analysis (contd.)

- A statistical test that leads to the conclusion that $\beta_1 \neq 0$ does not establish a cause-and-effect relation between the predictor and response variables, especially with non-experimental data. On the other hand, the existence of a regression relation in controlled experiments is often good evidence of a cause-and-effect relation.
- The confidence coefficients estimating a mean response $\hat{Y}_h \pm t(1 - \alpha/2; n - 2)s\{\hat{Y}_h\}$ and for the prediction limits $\hat{Y}_h \pm t(1 - \alpha/2; n - 2)s\{pred\}$ apply only for a single level of X .
- When observations on the predictor variable X are subject to measurement errors, the resulting parameter estimates are generally no longer unbiased.

Summary

- Inferences concerning β_1
- Inferences concerning β_0
- Interval estimation of $E\{Y_h\}$
- Prediction of new observation
- Analysis of variance approach
- Considerations in applying regression analysis

- 1 Chapter 2, Applied Linear Regression Models, Michael H. Kutner *et al.*, 2004